

QP Code : 14544

[3 Hours]

[Total Marks: 80

- N.B. (1) Question no. 1 is compulsory.
(2) Attempt any three from the remaining.
(3) Figures to the right indicate full marks.

1. (a) Find the Laplace Transform of $\sin t \cos 2t \cos ht$. 5
(b) Find the Fourier series expansion of $f(x) = x^2$ $(-\pi, \pi)$ 5
(c) Find the z-transform of $\left(\frac{1}{3}\right)^{k-1}$ 5
(d) Find the directional derivative of $4xz^2 + x^2yz$ at $(1, -2, -1)$ in the direction of $2\bar{i} - \bar{j} - 2\bar{k}$ 5
2. (a) Find an analytic function $f(z)$ whose real part is $e^x(x \cos y - y \sin y)$ 6
(b) Find inverse Laplace Transform by using convolution theorem $\frac{1}{(s-3)(s+4)^2}$ 6
(c) Prove that $\vec{F} = (6xy^2 - 2z^3)\bar{i} + (6x^2y + 2yz)\bar{j} + (y^2 - 6z^2x)\bar{k}$ is a conservative field. 8
Find the scalar potential ϕ such that $\nabla \phi = \vec{F}$. Hence find the workdone by $\vec{F}$ in displacing a particle from $A(1, 0, 2)$ to $B(0, 1, 1)$ along AB .
3. (a) Find the inverse z-transform of $F(z) = \frac{z^3}{(z-3)(z-2)^2}$ 6
(i) $2 < |z| < 3$ (ii) $|z| > 3$
(b) Find the image of the real axis under the transformation $w = \frac{2}{z+i}$ 6
(c) Obtain the Fourier series expansion of 8
 $f(x) = \pi x, 0 \leq x \leq 1$
 $= \pi(2-x); 1 \leq x \leq 2$
Here deduce That $\frac{1}{1^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{8}$
4. (a) Find the Laplace Transform of 6
 $f(t) = E; 0 \leq t \leq \frac{p}{2}$
 $= -E; \frac{p}{2} \leq t \leq p, f(t+p) = f(t)$

(b) Using Green's theorem evaluate $\int_C \frac{1}{y} dx + \frac{1}{x} dy$ where C is the boundary of the region bounded by $x=1$, $x=4$, $y=1$, $y=\sqrt{x}$

(c) Find the Fourier integral for $f(x) = 1-x^2$, $0 \leq x \leq 1$
 $= 0 \quad x > 1$

Hence evaluate $\int_0^{\infty} \frac{\lambda \cos \lambda - \sin \lambda}{\lambda^3} \cos\left(\frac{\lambda}{2}\right) d\lambda$

5. (a) If $\vec{F} = x^2 \vec{i} + (x-y) \vec{j} + (y+z) \vec{k}$ moves a particle from $A(1, 0, 1)$ to $B(2, 1, 2)$ along line AB . Find the workdone.

(b) Find the complex form of Fourier series $f(x) = \sin \pi x$ $(-l, l)$

(c) Solve the differential equation using Laplace Transform.
 $(D^2 + 2D + 5)y = e^{-t} \sin t$ $y(0) = 0$ $y'(0) = 1$

6. (a) If $\int_0^{\infty} e^{-2t} \sin(t+\alpha) \cos(t-\alpha) dt = \frac{3}{8}$ find the value of α .

(b) Evaluate $\iint_S (y^2 z^2 \vec{i} + z^2 x^2 \vec{j} + x^2 y^2 \vec{k}) \cdot \vec{n} ds$ where S is the hemisphere $x^2 + y^2 + z^2 = 1$ above xy -plane and bounded by this plane.

(c) Find Half range sine series for $f(x) = lx - x^2$ $(0, l)$

Hence prove that $\frac{1}{1^6} + \frac{1}{3^6} + \dots = \frac{\pi^6}{960}$